Analysis of Neural Models for Sandhi Splitting in Ayurvedic Sanskrit Texts

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY

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Outline

- Introduction
- Problem Definition
- Objectives
- Literature Review Summary
- Research Gap
- Dataset Description
- Methodology / Proposed Method
- Design / Architecture
- Evaluation Metrics
- Result Analysis
- Limitations
- Phase II plan
- Reference

Introduction

Ayurvedic Sanskrit texts such as the *Charaka Saṃhitā* and *Suśruta Saṃhitā* contain extensive medical knowledge accumulated over centuries. However, their computational processing is challenging due to linguistic complexities like sandhi, samāsa, rich morphology, and free word order. Recent advances in AI and NLP provide promising techniques for processing such classical texts. This project focuses on a evaluation of existing models applied to Ayurvedic Sanskrit for sandhi splitting.

Problem Definition

Ayurvedic Sanskrit texts are difficult to process computationally because of complex linguistic phenomena such as sandhi formation, compound words, and inflectional richness. Existing NLP models show varying performance. There is a need for a systematic approach for sandhi splitting involving Ayurvedic text.

Furthermore, there is a lack of comprehensive and dedicated models for Sandhi splitting Ayurvedic Sanskrit texts. The absence of such models leads to inconsistent and unreliable Sandhi splitting, particularly in complex compound formations and verse-level text.

Objectives

- To compare AI and NLP models for processing Ayurvedic Sanskrit texts
- To evaluate models on sandhi-splitting, morphological analysis, and NER
- To analyse strengths and limitations of rule-based, statistical, and transformer-based models
- To assess model performance using standard evaluation metrics
- To identify challenges specific to Sanskrit NLP
- To recommend suitable models for future Ayurvedic knowledge extraction systems

Literature Review Summary

Paper Title	Methodology	Dataset	Limitations
A Benchmark and Dataset for Post-OCR text correction in Sanskrit	Seq2seq correction baselines for Sanskrit OCR	218k Sanskrit OCR corrected sentences.	Needs Ayurveda specific adaptation. not domain focused
NLP-Based Knowledge Extraction from Charak Samhita for Navigating Ancient Wisdom	Django system: OCR + NER + query interface	OCR’d Charaka Saṃhitā	Used translations, Sanskrit-native pipeline needed
Digital Corpus of Sanskrit: Towards Improved Segmentation and Normalization	Normalization & segmentation improvements for Sanskrit	Digital Corpus of Sanskrit (DCS)	Lacks Ayurvedic corpora; calls for domain specific expansion
Building Named Entity Resources for Sanskrit Texts	Pre annotation + bilingual alignment + gazetteers	Śrīmad-Bhāgavatam subset (NER corpus)	Entities limited to persons/places. Ayurveda entities missing
Mapping Research Trends in Ayurveda: A Bibliometric Study (1993 2022)	Bibliometric mapping of Ayurveda research	11,773 Ayurveda publications	Not technical NLP; shows domain significance
Advances in Morphological Segmentation and Compound Analysis in Sanskrit	Morpho syntactic parsing for Sanskrit	Sanskrit Digital Corpus	No Ayurvedic integration; useful for preprocessing
Sanskrit Semantic Parsing for Knowledge Extraction	Semantic parsing for extracting relations	Sanskrit annotated datasets	Not domain specific; Ayurveda corpus missing

Research Gap

- Lack of integration of contemporary Pre-trained Language Models with Ayurveda

- Lack of domain-specific annotated corpora

- Inadequate handling of complex Sanskrit linguistic phenomena in domain context

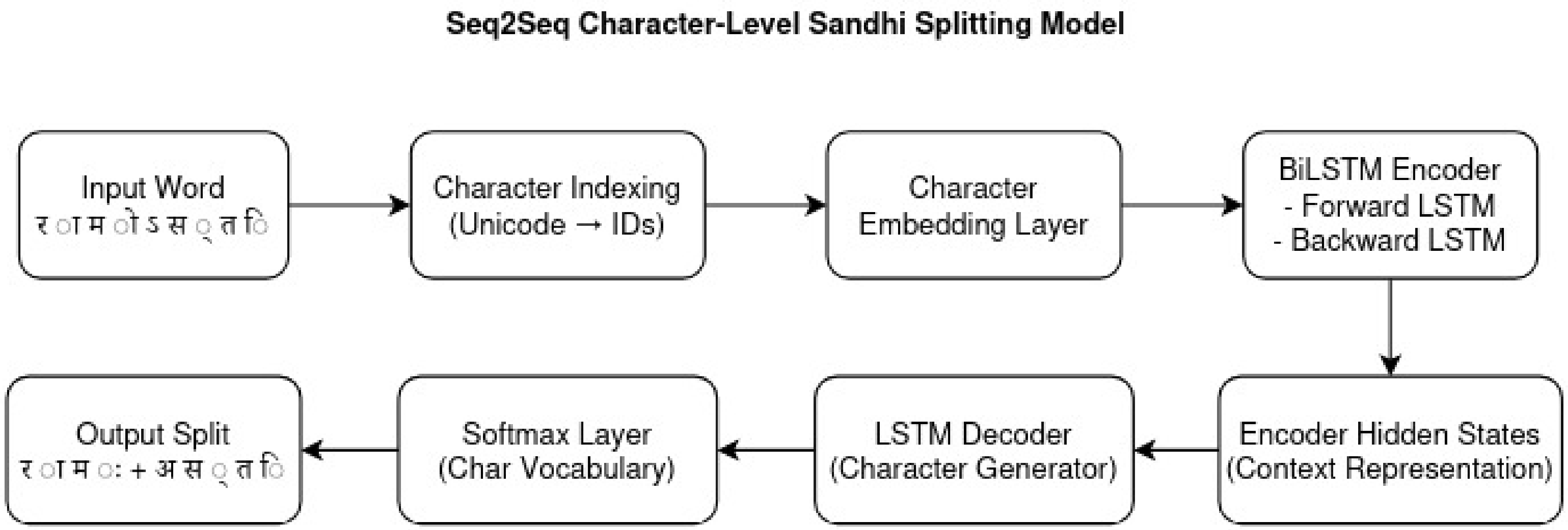
Dataset Description

- Source texts: Charaka Saṃhitā
- Source: University of Hyderabad
- Data type: Sanskrit verses with chapter markers
- Dataset size: Small, manually annotated

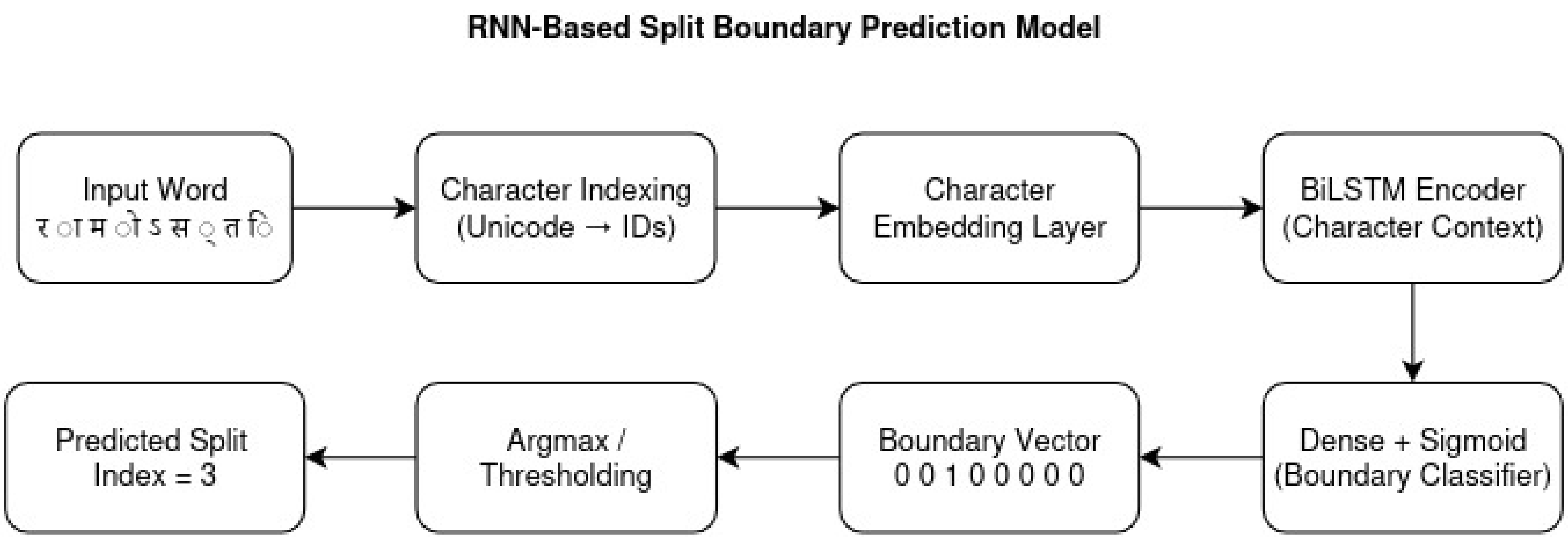
Methodology / Proposed Method

- Dataset Preparation – Selection and automated annotation of verses
- Preprocessing – Normalization, tokenization, cleaning
- Task Evaluation – Sandhi-splitting
- Evaluation of different models for Sandhi Splitting

Architecture - Seq2Seq

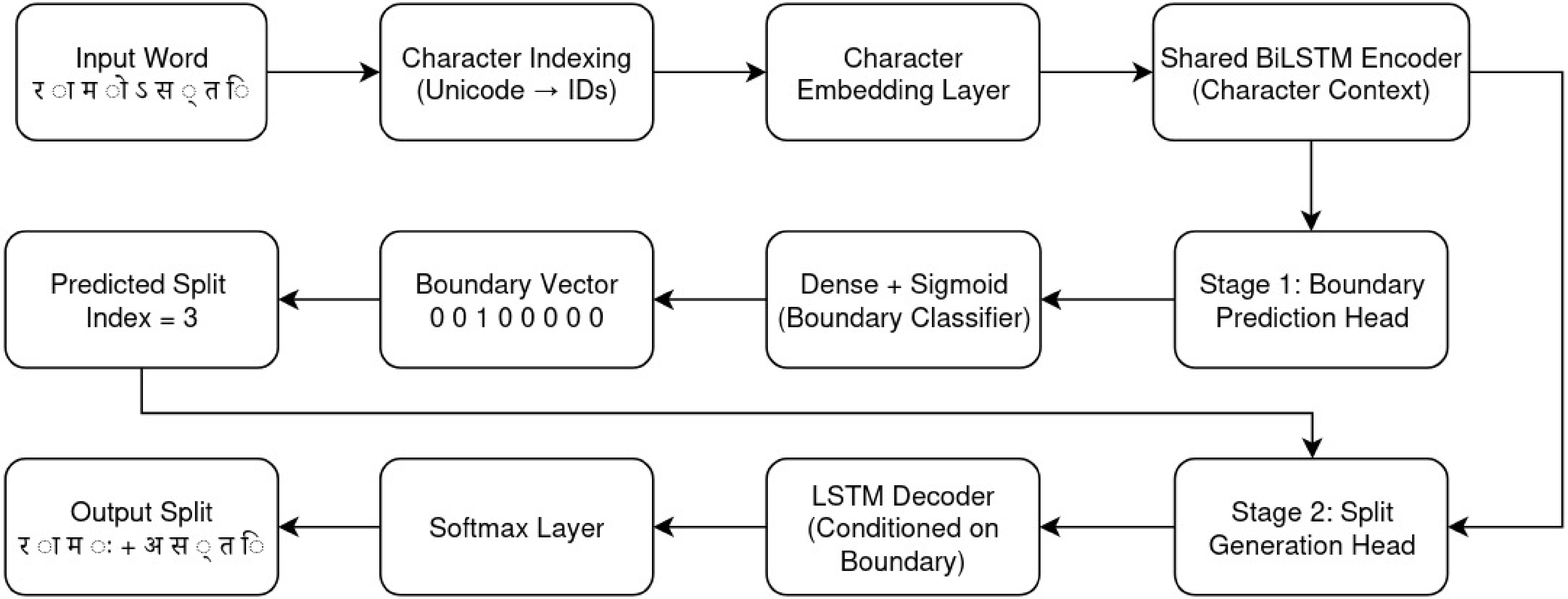


Architecture - RNN



Architecture - SandhiWindow

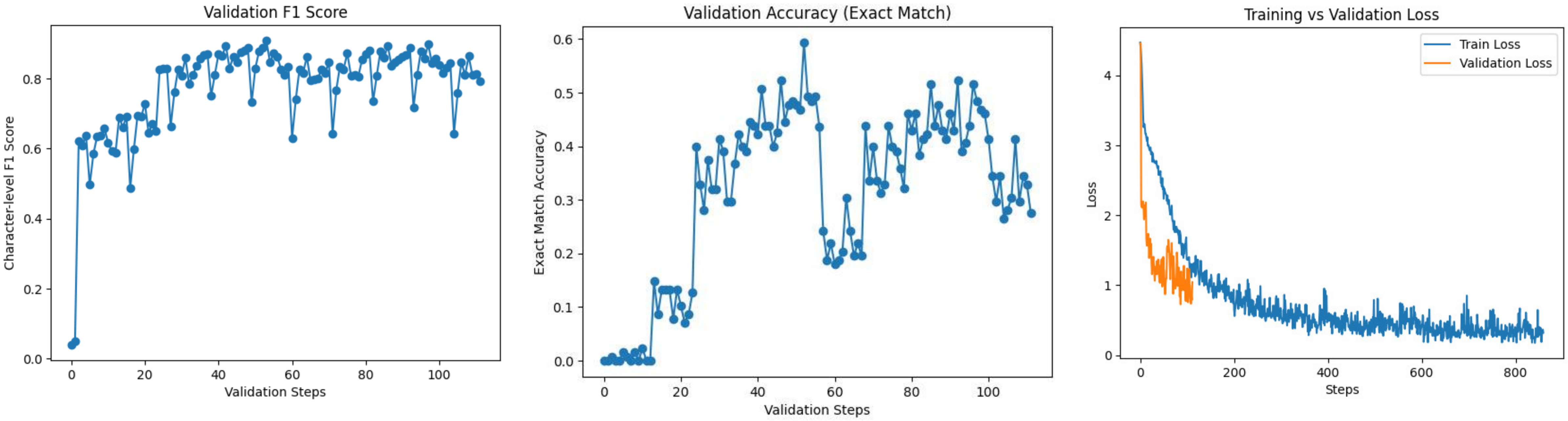
Two-Stage Neural Model for Sanskrit Sandhi Splitting



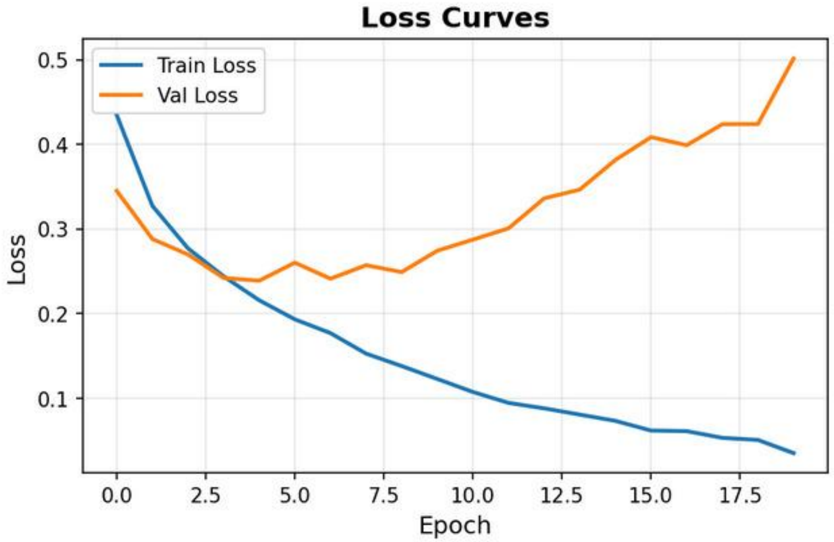
Evaluation Metrics

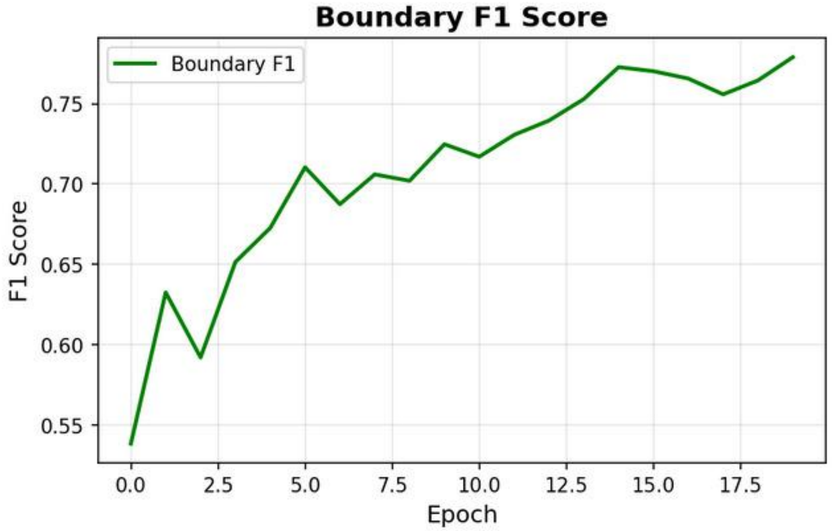
- Boundary F1-Score
  - Boundary F1 measures how accurately a model predicts the correct split boundaries in a word or sequence.
  - identify the correct boundary positions where a sandhi-joined word should be split.
  - Gold boundaries = correct split positions
  - Predicted boundaries = boundaries predicted by the model
- **Metrics:**
  - **Precision** $= \frac{\text{Correctly } predicted\ boundaries}{Total\ predicted\ boundaries}$
  - **Recall** $= \frac{\text{Correctly } predicted\ boundaries}{Total\ gold\ boundaries}$
  - **Boundary F1** $= 2 \times \frac{Precision \times Recall}{Precision + Recall}$

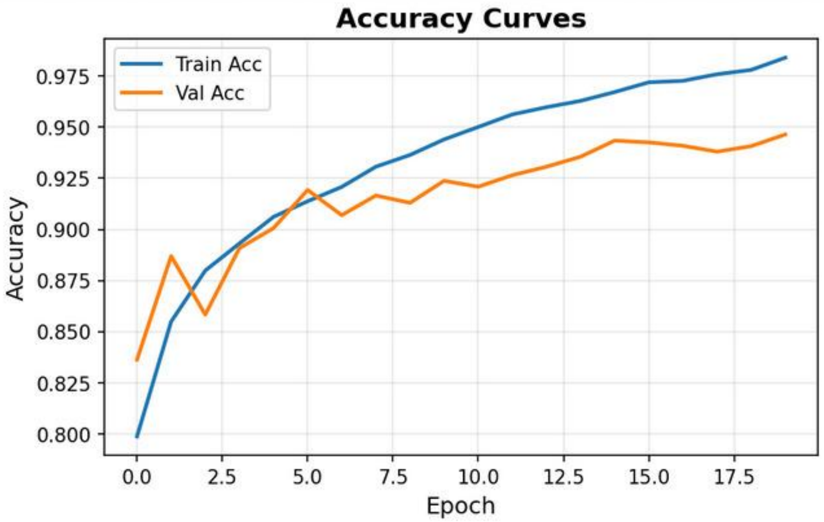
Result Analysis - Seq2Seq

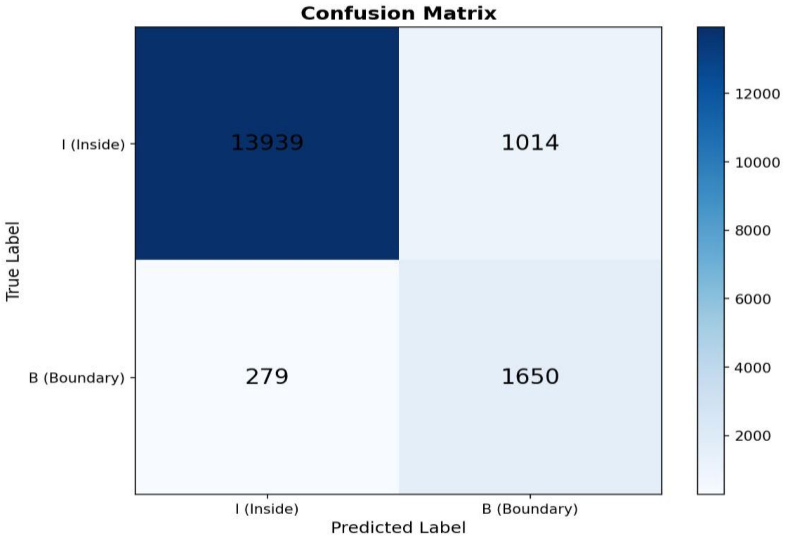


Result Analysis - RNN

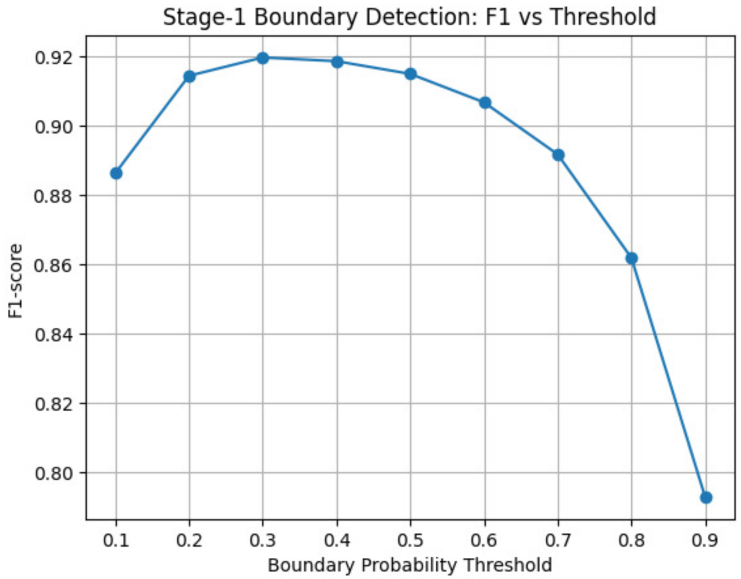


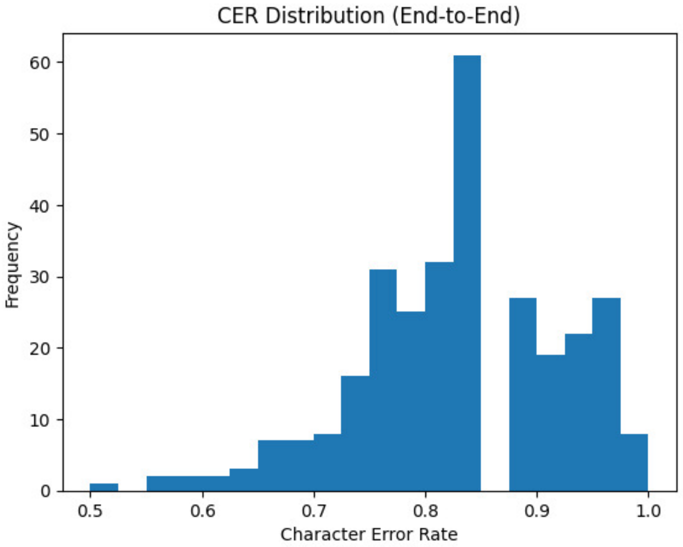


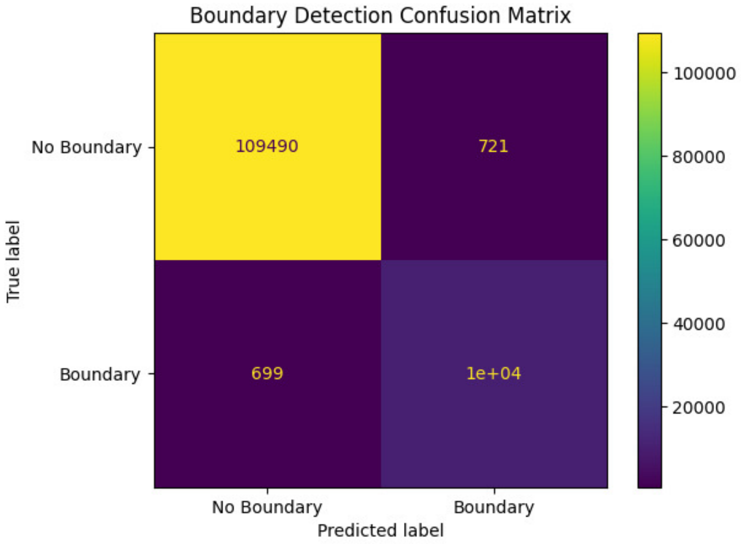


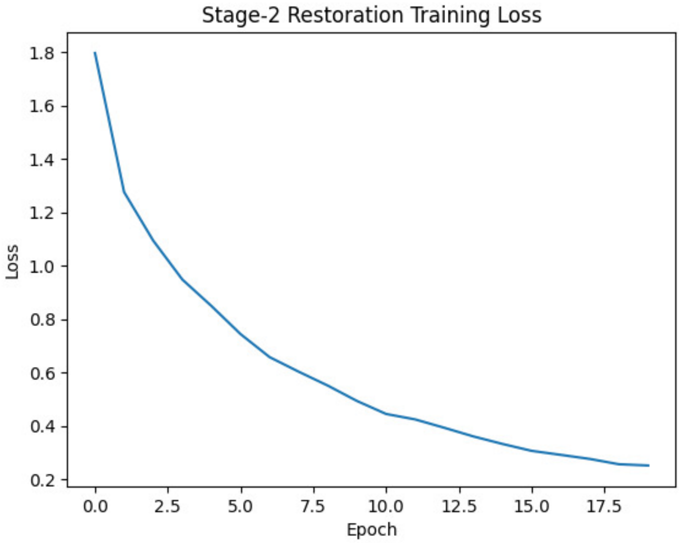


Result Analysis - SandhiWindow









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Limitations

- **Handling of Complex Sandhi and Compounds**
  Long compound words (*samāsa*) and complex sandhi formations remain challenging, leading to segmentation and interpretation errors.

- **Lack of Relation Extraction**
  The current work focuses primarily on segmentation, morphology, and entity recognition, without extracting semantic relations between entities.

- **Limited Annotated Dataset**
  The models were trained on a relatively small, manually annotated dataset, which restricts generalization to larger and more diverse Ayurvedic texts.

Phase II Plan (Proposed)

- **Fine-Tuning Domain-Specific Language Models**
  Further fine-tune advanced Sanskrit pre-trained language models specifically on Ayurvedic corpora to improve performance on medical entities and relations.

- **Hybrid NLP Approaches**
  Explore hybrid models that combine rule-based linguistic knowledge with transformer-based deep learning to better handle sandhi, long compounds, and free word order.

- **Clinical and Educational Applications**
  Apply the extracted structured knowledge to clinical decision support systems, digital textbooks, and intelligent Ayurvedic search engines.

Reference

- Aralikatte, R., Gantayat, N., Panwar, N., Sankaran, A., & Mani, S. (2019). *Sanskrit sandhi splitting using seq2(seq)²*.
- Hellwig, O. (2015). *Using recurrent neural networks for joint compound splitting and Sandhi resolution in Sanskrit*.
- Dave, S., Singh, A. K., P., P. A., & Lall, B. (2020). *Neural compound-word (Sandhi) generation and splitting in Sanskrit language*